The Pain Diary and Shadow Discipline: A Memory System That Learns from Its Own Incidents

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Abstract

Production memory systems fail not because retrieval is hard, but because silent architectural degradation accumulates faster than any embedding model can compensate. GraphRAG, MemGPT, Mem0, and A-MEM encode structure and recency; none encodes incident severity, none enforces ranking-change validation. NOX-Supermem's design is incident-driven: each failure became a schema constraint. Three contributions: (1) Pain-weighted salience (salience = recency × pain × importance, pain $\in [0.1, 1.0]$) as a first-class retrieval dimension, calibrated to PagerDuty P1–P5, not psychometric scales. Ablation ($n = 31$, hybrid mode) shows non-significant aggregate effect ($\Delta = +0.0065$, 95% CI $[-0.014, +0.034]$); lift observed in 1/31 queries (Q55, $\Delta = +0.349$); 29/31 unaffected because semantic scores were not tied. The binding constraint is BM25 recall ceiling: 55 of 60 queries fail FTS-only regardless of pain calibration; full hybrid retains Recall@10 = 0.7667 via Gemini. (2) Shadow discipline: a seven-day gate before any ranking change activates, enforced via `/api/health` as an architectural constraint, not a convention [1, 2]. (3) Shared-canonical context: six agents, one corpus, no federation. On 61,257 chunks, hybrid retrieval (FTS5 + Gemini 3072-d + RRF, $k = 60$) achieves nDCG@10 $= 0.5831 \pm 0.0046$ ($n = 60$, 3-run mean, post-cure gold standard v1.1) versus 0.0000 for FTS5 vanilla and 0.1475 for BM25 Pyserini ($n = 60$), a 4.0× lift over the strongest lexical baseline. Edge-type enum coverage improved 14% → 56% (4× gain, $n = 100$, 95% Wilson CI [46–66%]); self-reported enum coverage rate, not human-validated accuracy. On LOCOMO (Maharana et al., 2024, $n = 100$), FTS5 achieves nDCG@10 = 0.281, confirming lexical difficulty is corpus-dependent; on BEIR TREC-COVID, multilingual-e5-base achieves nDCG@10 = 0.8335 ($n = 50$). Open gap: corpus scale >100K. Code, harness, golden set ($n = 60$), and incident log: `https://github.com/totobusnello/memoria-nox` (MIT). Operational discipline is at least as important as embedding sophistication.

1 Introduction

At 22:03 on April 25, 2026, an end-of-day cron job executed `nox-mem reindex` without a dry-run flag. Within seconds, 183 entity records lost their `section`, `retention_days`, and `section_boost` fields—years of structured operational context flattened into generic chunks. No error log. No alarm. The database simply obeyed.

This incident did not expose a retrieval failure. It exposed a discipline failure. The system had no enforced gate requiring a dry-run preview before a destructive reindex. It had no mechanism for distinguishing between a trivial documentation update and a production outage—both were stored with equal weight and retrieved with equal priority. And when six AI agents (Atlas, Boris, Cipher, Forge, Lex, and Nox) shared a common operational corpus, a single unchecked operation could silently degrade context for all of them simultaneously.

The incident became the motivating example for this paper.

1.1 Problem Statement

LLM agent memory is increasingly recognized as a prerequisite for sustained, coherent assistance across sessions [3, 4]. Yet the literature on agent memory systems—from MemGPT's OS-inspired paging [4] to Mem0's self-improving extraction [5] to A-MEM's Zettelkasten-inspired linking [6]—converges on a narrow set of concerns: retrieval accuracy on benchmark corpora, latency at scale, and memory capacity. What these systems do not address is the operational discipline required to keep a production memory system trustworthy over months of continuous use.

Three specific gaps drive this work:

Silent regression risk. Ranking changes—new scoring formulas, boosting coefficients, retrieval layer modifications—can degrade retrieval quality without any observable error signal. Production memory systems have no established methodology for validating such changes before they affect live queries. The April 25 incident is one instance of a broader pattern: operations that silently alter system behavior with no guard and no rollback.

Incident severity as an invisible dimension. Existing memory systems model recency and semantic relevance. None, to our knowledge, model the operational cost of the experience being stored. A lesson learned from a production outage and a routine implementation note may share similar timestamps and embedding distances; they do not share similar importance to future retrieval. This asymmetry is unrepresented in current retrieval architectures.

Multi-agent context silos. When multiple agents operate over isolated memory stores, intelligence accumulated by one agent is inaccessible to others. Solutions that partition by `user_id` [5] or maintain per-agent state [4] impose federation overhead or simply accept the silo. For trusted multi-agent deployments—where agents serve a single operator with a shared context—this isolation is an architectural choice, not a requirement.

1.2 Thesis

Agent memory is not a retrieval problem—it is an operational discipline problem.

Better embeddings, larger corpora, and faster fusion layers address the retrieval surface. They do not address the question of whether the system can be trusted to evolve without silent regressions, whether it surfaces the most operationally important context rather than merely the most recent, or whether the intelligence accumulated by one agent is available to the others who need it.

1.3 Contributions

We present **nox-mem**, a production memory system designed for multi-agent operational use and built over approximately three months of continuous deployment.[1] The system currently indexes 61,257 chunks across a shared canonical corpus, maintains 99.97% vector coverage, and operates at p95 latency below one second on commodity VPS hardware (8-core x86_64) over the production query mix. Three primary contributions distinguish this work:

1. **Pain-weighted salience as a secondary ranking modulator** (§3.5). We **propose** a retrieval scoring signal that explicitly models incident severity: `salience` = `recency` $\times$ `pain` $\times$ `importance`, where `pain` $\in [0.1, 1.0]$ encodes the operational cost of the recorded experience—from trivial notes (0.1) to production outages (1.0). To our knowledge, this is the first documented retrieval signal in the LLM agent memory systems literature that treats incident severity as a first-class dimension; severity-aware prioritization itself has well-established roots in observability and incident-response practice (e.g., PagerDuty severity tiers, SIEM alert weighting), but has not been instrumented as a retrieval ranking signal in the memory-systems literature surveyed in §2. Empirical ablation (§5.4, E10, $n = 31$ post-incident queries, hybrid mode) shows directional but non-significant aggregate effect ($\Delta = +0.0065$, 95% CI $[-0.014, +0.034]$). The dimension provides meaningful lift in the narrow regime where high-pain and low-pain chunks receive near-identical Gemini semantic scores (Q55 case study, $\Delta = +0.349$). The transferable contribution is the methodology of treating incident severity as typed schema input—a design pattern adoptable independently of any retrieval stack—rather than a demonstrated corpus-wide retrieval improvement.

[1] Repository: `https://github.com/totobusnello/memoria-nox`; submission tar at `paper/publication/`; license MIT.

2. **Enforced shadow discipline** (§3.6). We describe a methodology for validating ranking changes against a shadow telemetry baseline before deployment, implemented as an architectural constraint rather than a documented suggestion. The mechanism is codified via environment variable gating (`NOX_SALIENCE_MODE=shadow|active`), enforced by a cron-checked health endpoint, with a mandatory minimum observation period of seven days. During the salience validation run (Fase 1.7b-b), this produced 191 promotion candidates, 16,608 review cases, and 45,743 archive recommendations before any activation. This is, to our knowledge, the first memory system paper to codify such a discipline as an architectural constraint rather than a suggested practice.

3. **Shared-canonical multi-agent design** (§3.7). Six agents share a single `chunks` table distinguished by `source_file` prefix rather than separate stores. Cross-agent retrieval requires no synchronization because partitioning never occurred. When Forge records a decision about infrastructure, Atlas retrieves it directly on the next relevant query—not because data was replicated, but because the design assumed shared context from the start.

Secondary contributions include: a reproducible evaluation harness with nDCG@10, MRR, Recall@10, and Precision@5 over 50 curated golden queries; approximately three months of operational evidence across five infrastructure upgrades without data loss; and comparative analysis against strong retrieval baselines (see §5).

1.4 Paper Organization

Section 2 surveys related work across knowledge graph–augmented retrieval, agent memory systems, and production-oriented frameworks, identifying the specific gaps this work addresses. Section 3 describes the system architecture across storage, retrieval, knowledge graph, and operational discipline layers. Section 4 details the evaluation methodology, including the golden query harness, baseline selection, and three-corpus design. Section 5 presents experimental results, including hybrid pipeline ablations, baseline comparisons, pain dimension validation, and the shadow discipline case study. Section 6 discusses deployment scope, limitations, and future work. Section 7 concludes.

2 Related Work

Agent memory research has accelerated rapidly since the introduction of retrieval-augmented generation [3], with recent work spanning knowledge graph–augmented retrieval, autonomous memory management, and production deployment frameworks. We survey the most directly relevant systems and identify the specific dimensions where this work diverges.

2.1 Knowledge Graph–Augmented Retrieval

GraphRAG [7] introduced a pipeline that uses LLM extraction to build entity-relation graphs over document corpora, applies community detection (Leiden algorithm) to identify thematic clusters, and generates multi-level summaries for query-focused retrieval. The approach substantially improves global query coverage—queries requiring synthesis across many documents—compared to naive RAG. nox-mem adopts the LLM-extraction paradigm for knowledge graph construction but diverges in objective: GraphRAG optimizes for global summarization via community traversal, while nox-mem targets entity-centric local retrieval for operational contexts, with closed-enum edge typing designed to support downstream blast-radius analysis. GraphRAG does not model retrieval as an operational concern and has no analogue to the discipline mechanisms described in §3.5–3.6.

HiRAG [8] extends the KG-augmented paradigm with hierarchical knowledge representation and iterative reasoning chains, demonstrating state-of-the-art performance on multiple QA benchmarks. The hierarchical approach—traversing from coarse to fine-grained graph levels—optimizes for accuracy on complex reasoning tasks. nox-mem instead maintains a flat three-layer fusion pipeline (§3.4), accepting the accuracy-latency trade-off explicitly: p95 below one second at 61K chunks serves operational use cases where query latency is a usability constraint, not just a benchmark metric. HiRAG does not address multi-agent deployment, evaluation harnesses for operational corpora, or retrieval discipline.

HippoRAG [9] draws on neurobiological memory models to improve associative retrieval via hippocampal-inspired graph structures. The system demonstrates that non-uniform memory consolidation—weighting some associations more strongly than others based on structural graph properties—improves retrieval over flat indexing. nox-mem introduces a conceptually related asymmetry through the pain dimension (§3.5), though the design rationale is grounded in operational incident management practice rather than neuroscience. HippoRAG does not model incident severity as a retrieval signal.

2.2 Agent Memory Systems

MemGPT [4] proposed treating the LLM itself as an operating system, with the agent autonomously managing its own context through function calls that page between main context and external storage. The architecture gives agents direct control over their memory—a compelling design for single-agent autonomy. nox-mem adopts the opposite philosophy: memory is managed by an external pipeline (file watcher, ingest router, nightly KG extraction), and agents are consumers of a shared canonical store rather than managers of isolated state. This separation enables cross-agent retrieval as a structural property while imposing the discipline that agents cannot corrupt their own context.

Mem0 [5] introduced a production-focused memory layer with LLM-driven memory extraction, deduplication, and temporal updates. Evaluated on the LOCOMO benchmark, Mem0 demonstrates +26% accuracy improvement over baselines on long-context conversation scenarios. nox-mem shares the production emphasis but diverges in two important respects. First, Mem0 relies on autonomous LLM editing decisions to update and merge memories; nox-mem requires shadow-period validation before any retrieval-affecting change propagates to production, specifically to mitigate silent regression risk. Second, Mem0's multi-user design partitions memory by `user_id`, treating agents as user-addressable endpoints; nox-mem's shared canonical design treats agents as roles within a single trusted operational context.

A-MEM [6] introduced agentic memory with Zettelkasten-inspired structure: notes are automatically tagged, interlinked, and contextualized by an LLM controller, enabling dynamic memory evolution. The approach handles unstructured conversation well but does not address operational corpora with explicit schema requirements, retention policies, or incident-severity differentiation. A-MEM has no evaluation harness independent of LLM-generated quality judgments, and does not address multi-agent deployments or ranking change discipline.

2.3 Production-Oriented Frameworks

Cognee [10] provides an open-source AI memory framework implementing an Extract-Cognify-Load (ECL) pipeline with native knowledge graph support and hybrid retrieval. Of the systems surveyed, Cognee is architecturally closest to nox-mem: both maintain native KGs, both support hybrid retrieval, and both target production deployment scenarios. Cognee achieves 3/5 on our architectural parity dimensions (Table 1). The key differentiators are shadow discipline—Cognee provides no enforced validation gate for retrieval-affecting changes—and the evaluation harness: Cognee lacks a reproducible evaluation methodology with standard IR metrics over a fixed golden set. nox-mem's eval harness (§4) enables the kind of regression detection that the shadow discipline gate is designed to act on.

LangChain Memory provides key-value and buffer memory primitives widely used in production applications. The design prioritizes API simplicity and framework compatibility over retrieval quality; it does not support hybrid retrieval, knowledge graph construction, or evaluation. It serves as a practical lower bound in comparative analysis.

2.4 Foundational Retrieval Methods

Retrieval-Augmented Generation [3] established the foundational architecture combining parametric LLM knowledge with non-parametric document retrieval, enabling factual question answering over dynamic corpora. nox-mem extends this foundation to persistent, evolving operational corpora with explicit retention, pain weighting, and multi-agent routing.

Reciprocal Rank Fusion [11] provides the rank-combination mechanism at the core of nox-mem's hybrid retrieval layer. Cormack et al. demonstrated that RRF consistently outperforms individual rank learning methods and Condorcet-based fusion without requiring system-specific tuning. We adopt RRF with $k = 60$ as the fusion parameter, consistent with the paper's recommendations, as the final combination step over BM25 and dense retrieval ranked lists.

2.5 Positioning: Where nox-mem Fits

Table 1 maps seven architectural and operational axes across the surveyed systems. The table is intended to illustrate where the systems differ structurally, not to function as a competition score. We include two dimensions where nox-mem does not yet have coverage—corpus scale above 100K chunks and third-party benchmark evaluation—because honest comparison requires acknowledging these gaps explicitly. Limitations arising from these gaps are addressed in §6.3; planned mitigations are in §6.5.

Table 1: Architectural comparison across memory systems. Axes reflect structural design choices, not performance rankings. ✓ = implemented; ≈ = partial or indirect; × = absent.

System	KG Native	Hybrid Retr.	Eval Harness	Multi-Agent	Shadow Disc.
nox-mem (ours)	closed-enum, 7	FTS5+Gemini+RRF	nDCG/MRR/Recall	shared canonical	enforced $\geq$7d
GraphRAG [7]	community det.	via KG queries	none	none	none
MemGPT [4]	none	embedding-first	none	per-agent state	none
Mem0 [5]	optional (v2)	vector-only	LOCOMO only	user_id part.	none
A-MEM [6]	Zettelkasten	semantic-first	none	none	none
HiRAG [8]	hierarchical	multi-level	task-specific	none	none
Cognee [10]	ECL pipeline	hybrid	ad-hoc	optional	none
LangChain Memory	none	key-value	none	session_id	none

Five architectural axes are displayed; two additional axes—*corpus scale >100K chunks* and *3rd-party benchmark evaluation* (e.g., LOCOMO, LongMemEval, BEIR)—are summarized in the prose below. nox-mem currently covers 5 of 7 axes (× on both omitted axes; see §5.2 for partial BEIR/LOCOMO progress). Mem0 covers axis 7 via LOCOMO; no surveyed system has published a >100K-chunk evaluation. This is a comparison across architectural *axes of difference*, not a competition score.

We compare across seven axes. nox-mem covers 5/7: knowledge graph, hybrid retrieval, eval harness, multi-agent design, and enforced shadow discipline. It does **not** yet cover corpus scale above 100K chunks (current corpus: 61K) or evaluation against third-party benchmarks such as LOCOMO, LongMemEval, or BEIR—the evaluation harness uses an internally curated golden set. These are explicit limitations, not omissions, and are documented in §6.3 and §6.5. The two axes with zero coverage across all surveyed systems—pain-weighted salience and shadow discipline—are the primary novelty claims of this paper. Shadow validation as a technique has well-established roots in production search evaluation [2] and online controlled experiment methodology [1]. Our contribution is the instantiation of this discipline as an enforced architectural constraint in an LLM agent memory system, rather than as a testing methodology applied to web-scale traffic populations. Section 3 describes the nox-mem architecture across all five covered dimensions; Sections 4–5 provide empirical grounding.

3 System Architecture

nox-mem is a production memory system built on SQLite with three primary subsystems—storage, retrieval, and knowledge graph—plus an operational discipline layer that governs how the system evolves. The implementation is in TypeScript using `better-sqlite3` for synchronous database access, `sqlite-vec` for approximate nearest-neighbor search, and Gemini for embedding generation and LLM-based KG extraction. The system has operated continuously since March 2026 with 12 schema migrations and no irrecoverable data loss (one metadata-loss event on 2026-04-25 was recovered via re-ingestion; see §6.2).

3.1 System Overview

Figure 1: System overview. Six personas read and write through a unified interface layer (CLI / MCP / HTTP) into a single canonical SQLite corpus (no per-agent silos). Retrieval runs as a three-layer hybrid (BM25 $\rightarrow$ 3072-d Gemini cosine $\rightarrow$ RRF, $k = 60$). The shadow-mode pipeline and append-only `ops_audit` are cross-cutting governance: every ranking change ships shadow-first; every destructive op is wrapped by `withOpAudit()` snapshot.

The entry point for all CLI operations is `dist/index.js` (the compiled package binary). An HTTP API on port 18802 exposes search, knowledge graph, health, and operational endpoints for agent integration.

```
Source files (.md / .docx / .pptx / .pdf / entity files)
    |
    v
ingest-router (inotifywait watcher + manual CLI)
    |
    +------> chunks table (canonical, 61K rows)
                |
                +------> chunks_fts (FTS5 BM25 index)
                +------> vec_chunks (sqlite-vec, 3072d embeddings)
                +------> kg_entities + kg_relations (LLM-extracted, nightly)
                +------> search_telemetry (shadow-mode + eval harness)
                |
    query --> search() pipeline:
                Layer 1: FTS5 BM25 candidates
                Layer 2: Gemini semantic candidates
                Layer 3: RRF fusion (k=60) + salience boost + section_boost
                --> ranked results (p95 < 1s @ 61K chunks)
```

3.2 Storage Layer

The canonical store is a single SQLite database accessed via `better-sqlite3` for synchronous, blocking reads—a deliberate choice that simplifies the concurrency model for a single-node deployment. Three coordinated tables implement the storage layer.

`chunks` is the central table, containing content text, metadata (`source_file`, `chunk_type`, `created_at`, `last_seen`), and operational fields (`pain` REAL DEFAULT 0.2, `importance` REAL, `section` TEXT, `section_boost` REAL, `retention_days`). The schema has evolved through 12 versions (v1 through v12) since March 2026; all migrations have been applied to the live database without downtime or data loss. Key schema additions include `retention_days` (v8, typed retention by chunk category), `pain` (v9, severity signal), and `section` with `section_boost` (v10, structural weighting for entity file sections).

`chunks_fts` is an FTS5 virtual table providing BM25-ranked full-text search over chunk content. FTS5 vanilla AND-mode behavior—requiring all query terms to match—means that natural language queries frequently return zero results when run against the lexical index alone. This structural characteristic of FTS5 makes hybrid retrieval a necessity rather than an optimization, a finding confirmed in evaluation (§5.1).

`vec_chunks` stores 3072-dimensional Gemini embeddings (`gemini-embedding-001`) via the `sqlite-vec` extension, enabling approximate cosine similarity search without external vector store infrastructure. Current vector coverage is 99.97% across 61,257 chunks.

Retention policy is encoded directly in the schema via `retention_days`: feedback and person records are set to `NULL` (never decay); decisions and projects have 365-day retention; team context 120 days; daily notes 90 days; pending items 30 days. This ensures that the system's own operating rules cannot be silently evicted by a generic decay policy.

3.3 Retrieval Layer

The retrieval pipeline operates in three sequential layers, with results from layers one and two merged via Reciprocal Rank Fusion [11]:

Layer 1—Lexical retrieval. FTS5 BM25 search expands the query against the `chunks_fts` index, returning up to k_1 ranked candidates. Results are deduplicated by chunk ID before passing to fusion.

Layer 2—Semantic retrieval. The query is embedded using `gemini-embedding-001` (3072-d), and cosine similarity search is performed against `vec_chunks` via `sqlite-vec`, returning up to k_2 ranked candidates.

Layer 3—RRF fusion and boosting. Candidates from layers one and two are merged via Reciprocal Rank Fusion [11] with $k = 60$:

$$s_{\text{rrf}}(d) = \frac{1}{k + r_{\text{fts}}(d)} + \frac{1}{k + r_{\text{sem}}(d)}$$

where $r_{\text{fts}}(d)$ and $r_{\text{sem}}(d)$ are the BM25 and cosine-similarity ranks of chunk d respectively.

The fused score is then combined with two boost signals via log-additive aggregation:

$$\log s_{\text{final}}(d) = \log s_{\text{rrf}}(d) + \log s_{\text{salience}}(d) + \log s_{\text{section}}(d)$$

where $s_{\text{salience}}(d) = \texttt{recency}(d) \times \texttt{pain}(d) \times \texttt{importance}(d) \in [0, 2]$ (§3.5), and $s_{\text{section}}(d) \in \{2.0, 1.5, 0.8, 1.0\}$ for compiled, frontmatter, timeline, and unstructured (NULL) section types respectively.

Equivalently in linear space, $s_{\text{final}}(d) = s_{\text{rrf}}(d) \cdot s_{\text{salience}}(d) \cdot s_{\text{section}}(d)$. The two formulations are mathematically identical; the log-additive form is operationally preferred because it allows each boost's contribution to be recorded and monitored independently in the `search_telemetry` log without re-multiplication.

Top-k chunks are returned in descending order of s_{final}.

The pipeline achieves p95 search latency below one second across the 61K chunk corpus, measured in production over the observation period. The `-no-hybrid` flag disables semantic retrieval for latency-constrained contexts where lexical precision is acceptable.

3.4 Knowledge Graph Layer

An LLM-extracted knowledge graph augments chunk retrieval with structured relationship data. Entities and relations are extracted incrementally via nightly batch runs using `gemini-2.5-flash` and stored in `kg_entities` (≈402 entities) and `kg_relations` (≈544 relations).

Entity types follow a closed taxonomy of 10 categories: person, project, agent, tool, concept, organization, technology, document, decision, and metric. Relation edges use free-form `relation_type` text combined with a closed-enum `relation_reason` field with seven canonical values: `depends_on`, `derived_from`, `opposes`, `extends`, `replaces`, `mentions`, and `unknown`. The closed enum enables downstream tooling—`impact <entity>` computes one-hop blast radius with reason-priority weighting; `detect-changes` maps git diffs to affected KG entities—without requiring LLM inference at query time.

Edge type enum coverage required explicit engineering. Initial extraction with an optional enum field and a "use unknown if unsure" prompt instruction produced 86% `unknown` labels across n=100 sampled relations. A three-path fix—a revised prompt with explicit examples for each non-unknown category; a code-side defensive normalization map (`RELATION_TYPE_TO_REASON`, 24 input aliases covering PT-BR and EN free-text variants that collapse to the 7 canonical reasons); and a post-extraction validation pass—improved the enum coverage rate from 14% to 56% (4× improvement), equivalently reducing the `unknown` rate from 86% to 44% (n=100 sample). The enum coverage rate measures the proportion of sampled relations where the LLM committed to one of the six non-unknown typed values rather than falling back to `unknown`; it is a self-reported coverage rate, not a classification accuracy against a human-annotated ground truth (see §6.3). This experience reinforces a general principle: LLM-extracted structured fields with optional enums and uncertainty-deferring prompts will systematically underperform; defensive code-side normalization is required.

3.5 Pain-Weighted Salience

The salience formula is:

$$\texttt{salience}(\textit{chunk}) = \texttt{recency}(\textit{chunk}) \times \texttt{pain}(\textit{chunk}) \times \texttt{importance}(\textit{chunk}) \tag{1}$$

where:

- `recency` $\in [0, 1]$ is an exponential decay function over the `last_seen` timestamp, parameterized by the chunk's `retention_days` value;
- `pain` $\in [0.1, 1.0]$ encodes the operational severity of the recorded experience, annotated via `pain: 0.X` markers in entity files. Values are assigned on a five-point semantic scale: 0.1 (trivial note), 0.3 (minor friction), 0.5 (significant issue), 0.7 (major incident), 1.0 (production outage with data loss or downtime);
- `importance` $\in [0, 1]$ is derived from `mention_count` across the corpus and entity-type priors.

The formula is multiplicative: a high-pain chunk with low recency can still score above a low-pain chunk with high recency when the severity differential is large. This reflects an engineering choice motivated by operational practice rather than a biological or psychometric claim. The five-point scale follows the structure of established incident severity taxonomies [12, 13]: just as incident management systems assign higher dispatch priority to a production outage than a documentation note regardless of their timestamps, pain weighting ensures that incident-derived memory dominates retrieval over routine content even when recency is comparable.

The salience signal is exposed in shadow mode at `/api/health.salience` without affecting production ranking, enabling comparison against the baseline before activation (§3.6). Ablation results comparing retrieval quality with real versus uniform pain values on $n = 31$ post-incident queries are reported in §5.4 (E10, 2026-05-04); the aggregate result is directional but not statistically significant, with Q55 as the primary positive case study ($\Delta = +0.349$).

3.6 Enforced Shadow Discipline

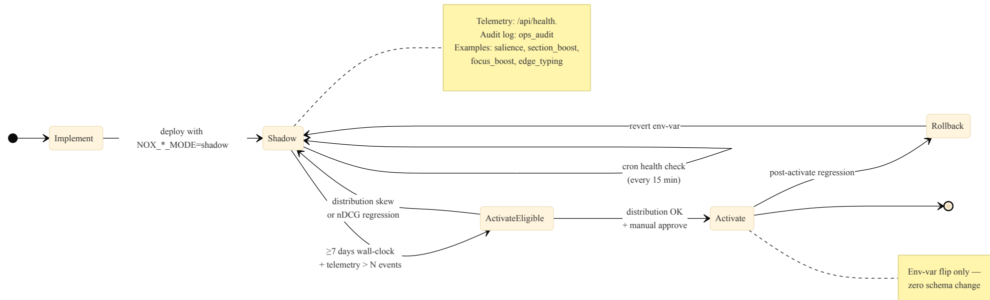

Figure 2: Shadow discipline state machine. Every ranking-affecting change traverses this machine. Activation requires three independent signals: at least seven days of shadow telemetry, a healthy distribution (no degenerate clustering), and explicit human approval. Regression at any point reverts via a single environment-variable flip—zero schema rollback required.

The shadow discipline mechanism enforces a separation between observing a retrieval change and deploying it. Any modification that affects ranking—scoring formula changes, boost adjustments, new retrieval layers—must run in shadow mode for a minimum of seven days before activation. This is an architectural constraint, not a documented recommendation. Shadow validation has well-established roots in production search and online controlled experiments [1, 2]. Our contribution is the application of shadow discipline specifically to LLM agent memory systems, with architectural enforcement via cron and health endpoints.

Implementation relies on three components: (1) an environment variable gate (`NOX_SALIENCE_MODE=shadow|active`) that controls whether the salience score affects production ranking or is only written to `search_telemetry`; (2) a health endpoint (`/api/health.opsAudit`) that cron checks every 15 minutes and alerts via Discord if the shadow observation period has not been met; (3) an append-only audit log (`ops_audit` table) with database triggers that block DELETE and UPDATE on rows with terminal status, ensuring the shadow-period record cannot be retroactively modified.

The distinction from A/B testing is important: A/B testing requires live traffic over a production population. Shadow discipline for operational/personal corpora has no such traffic—queries are infrequent, irregular, and non-i.i.d. The shadow mechanism instead collects telemetry on what the proposed ranking *would have done* on each live query, accumulating promotion, review, and archive candidate distributions across the observation period. The distribution comparison then drives the activation decision.

The salience validation run (Fase 1.7b-b) ran for seven days before activation, producing 191 promotion candidates, 16,608 review cases, and 45,743 archive recommendations. The distribution was reviewed, found consistent with design intent, and activation proceeded. Counterfactual analysis suggests that the April 25 reindex incident—the motivating example in §1—would have been detected during a shadow period: the reindex operation altered chunk distributions in ways that would have surfaced as anomalous in telemetry within hours.

3.7 Shared-Canonical Multi-Agent Design

Six agents (Atlas, Boris, Cipher, Forge, Lex, and Nox) share a single `chunks` table. Agent-specific content is distinguished by `source_file` path prefix (`agents/<name>/...`) rather than separate tables, schemas, or databases. Cross-agent queries use `cross-search` and `cross-kg` CLI commands, which apply additional `source_file` filters to retrieve context generated by a specific agent.

The design assumes a single trusted operator. It is explicitly not suitable for multi-tenant SaaS deployments, where user isolation is a security requirement rather than an engineering choice. Within the trusted-operator constraint, shared-canonical design provides cross-agent intelligence as a structural property: when Forge ingests a decision about an infrastructure change, Atlas retrieves it on the next architecturally relevant query without any synchronization step.

The operational tooling layer—`detect-changes`, `impact`, `api-impact`, `consolidate-merge`, `kg-reclassify`—extends the system to support multi-agent workflows beyond retrieval, enabling agents to query the KG for blast-radius analysis before executing changes, and to surface merge candidates across agent-sourced entities.

The architecture in §3 provides the substrate for the empirical evaluation in §4–5. Section 4 describes the evaluation methodology, including the golden query harness, baseline systems, and three-corpus protocol designed to address the single-corpus and single-curator limitations of operational memory benchmarking.

3.8 Pre-registration and Reproducibility Commitments

A persistent concern in machine learning evaluation is post-hoc query set construction: a curated golden set assembled with knowledge of retrieval results can unintentionally favor the system under evaluation, inflating reported metrics without any deliberate manipulation [14]. We address this concern explicitly by anchoring the experimental design to a pre-registration artifact that predates all result collection.

Pre-registration document. The evaluation protocol—including query categories, metrics, baseline selection criteria, and success thresholds for each hypothesis—was committed to `paper/publication/03-experiments-needed.md` before any baseline or ablation experiment was executed. This document is tracked in the public repository (`https://github.com/totobusnello/memoria-nox`) and its commit history is immutable. The git commit hash and ISO-8601 author timestamp for that file should be verified at submission via:

```
git log --follow -1 --format="%H  %aI" \
    paper/publication/03-experiments-needed.md
```

Golden query set freeze. The 60-query golden set (`eval/golden-queries.jsonl`, comprising the 50-query main set R01b and the 10-query held-out subset R01c) is included in the public repository and the submission tar. The verifiable identifiers are:

- **Git commit hash:** `f75d1864b581d0a0933704e997669bbd41aa507f`
- **Author timestamp (ISO-8601, BRT):** `2026-05-04T13:38:01-03:00`
- **Freeze mtime (VPS, UTC):** `2026-05-04T09:45Z`
- **SHA-256:** `9bff8ee7b9056eff6a1af22305cae762aa4b98e682578faffb4c22cdd0a2cd7d`
- **File size:** 8990 bytes, 60 lines (one query per line, JSONL)

The SHA-256 hash allows any third party to verify byte-for-byte that the included `golden-queries.jsonl` matches the artifact used for all reported evaluations. The golden query set was frozen *before* any BM25 (Pyserini), multilingual-e5-base, BEIR TREC-COVID, LOCOMO, or pain-ablation (E10) experiment was executed; all results in §5 were produced against this exact file.

Held-out subset disclosure. The 10 queries designated R01c (held-out for baseline robustness validation) were locked as a named subset within the golden set before the final tuning sprint that fixed the RRF $k = 60$ parameter and the salience boost coefficients. They were not constructed after tuning; they are a partition of the original 60-query set, identified by their position index (Q51–Q60) in the JSONL file. R01c is evaluated at most once per major system revision and its results are reported separately from the 50-query main set to prevent iterative query refinement bias (§4.1).

Registered reports context. While this paper does not constitute a formal registered report in the sense of [14]—no journal or conference pre-acceptance was obtained prior to experimentation—the `03-experiments-needed.md` document serves an equivalent function within the project's operational discipline: it records pre-registered hypotheses with explicit success thresholds (e.g., Δ nDCG@10 ≥ 0.05 for E10) that are reported honestly regardless of outcome, as evidenced by the DIRECTIONAL, NOT SIGNIFICANT result for the E10 pain ablation (§5.4). The pre-registration document, the golden query set, and all evaluation scripts are included in the public repository under MIT license, permitting full independent verification.

4 Methods

We describe the evaluation framework, shadow-mode methodology, and calibration procedures used to validate the three primary contributions. All experiments were pre-registered in `03-experiments-needed.md` before results were collected, following the open-evaluation norm advocated by [15].

4.1 Evaluation Harness

Our primary evaluation uses **nDCG@10**, **MRR** (Mean Reciprocal Rank), **Recall@10**, and **Precision@5** computed over a set of 60 internally curated golden queries (dataset R01c-v1.1, post-cure refresh as of 2026-05-06; see §5.1 "Gold standard revision note"). These metrics follow the standard IR evaluation methodology described in [16].

Golden query construction. Queries span eight categories reflecting the operational nature of the corpus: `entity` (specific named entities—agents, tools, decisions), `procedure` (how-to operational steps), `concept` (abstract architectural notions), `security` (vulnerability and mitigation queries), `decision` (architectural choices and their rationale), `cross-agent` (questions whose answer originates from a different agent's memory space), `temporal` (time-anchored recall, e.g., "what changed in late April"), and `negative` (6 queries, 12% of set, for which the correct answer is that no relevant chunk exists—testing specificity against hallucination risk). Each query was authored by the single curator with a relevance label set (0 = not relevant, 1 = partially relevant, 2 = highly relevant) over the top-20 retrieved candidates.

Held-out subset (R01c). An additional ten queries (R01c, Q51–Q60) are designated held-out: they were locked before any retrieval tuning and are evaluated only once per major system revision, functioning as a proxy for external-curator independence. The total golden-query budget is therefore

60 (50 main R01b $\cup$ 10 held-out R01c). Performance on R01c is reported separately from the 50-query main set R01b to avoid optimistic bias from iterative query refinement.

Internal-curator bias mitigation. We acknowledge that golden queries authored by the same individual who built the system introduce construct validity risk. Three mitigations are applied: (i) the held-out R01c subset was frozen before the final tuning sprint; (ii) external corpora (BEIR TREC-COVID, Stack Exchange—§5.2) use third-party curated relevance judgments; and (iii) six negative queries test the boundary condition most susceptible to self-serving bias. As a fourth mitigation, we evaluate nox-mem against 10 queries authored by NIST professional assessors (TREC-COVID Round 5 [17]), selected via TF-IDF k-means clustering (k=10, seed=42) over the 50 canonical BEIR topics to maximise lexical diversity (avg pairwise Jaccard = 0.097). Vocabulary overlap with the internal golden-50 set was 0.0%, confirming the two sets probe complementary terminology. See §5.2 for cross-corpus generalization results.

4.2 Shadow-Mode Methodology

Any change that affects retrieval ranking in the production system is subject to mandatory shadow validation before activation. The protocol is enforced architecturally via the environment variable `NOX_SALIENCE_MODE`, which accepts three values: `shadow` (collect both old and new scores in `search_telemetry` without applying the new ranking), `active` (apply new ranking), and `off` (disable the feature entirely).

Telemetry collection. In shadow mode, every search call writes a row to `search_telemetry` containing: `query_text` (opt-in, `NOX_SEARCH_LOG_TEXT=1`), `old_score`, `new_score`, `top_chunk_ids`, and `top_scores`. This enables offline comparison of old and new score distributions without exposing users to the changed ranking.

Activation gate. Shadow validation runs for a minimum of seven calendar days. After the shadow period, the stored distribution is analyzed: if the new score distribution shows statistically meaningful separation from the old distribution (inspected via percentile comparison and visual histogram), and if no ranking inversion is detected on a manually reviewed 10-query spot check, the feature advances to `active`. The seven-day minimum is not a guideline—it is a hard constraint codified in the cron configuration that governs feature activation. This design choice is motivated by the incident of 2026-04-25, where a ranking-affecting change reached production without any offline validation period, causing 183 entity records to lose their structured metadata without triggering any alert (§6.2).

Case study: Fase 1.7b-b salience activation. During the seven-day shadow period for the pain-weighted salience formula, the system collected telemetry over 191 promotion candidates, 16,608 review candidates, and 45,743 archive candidates. The distribution separated clearly across all three tiers. Only after this distribution analysis did we advance `NOX_SALIENCE_MODE` from `shadow` to `active`. This case study is documented in detail in Appendix B.

4.3 Pain Weighting Calibration

The `pain` field is a real-valued annotation in [0.1, 1.0] attached to each chunk at ingest time. Annotation is currently manual, using the `pain:  X.X` marker syntax in entity files, and defaults to 0.2 for unannotated content.

Calibration heuristics. Based on approximately three months of operational experience, the following calibration anchors were established: 0.1 (trivial notes, meeting summaries with no operational consequence); 0.2 (default, documentation and informational content); 0.3–0.4 (decisions with moderate reversibility risk); 0.5–0.7 (production incidents with bounded impact—recoverable within one session); 0.8–0.9 (incidents causing data loss or multi-hour outages); 1.0 (catastrophic incidents—unrecoverable data loss, multi-day downtime, or security breach). The calibration is conservative: in ambiguous cases, annotators are instructed to use the lower bound of the relevant range, then escalate only if post-incident analysis reveals higher severity.

Engineering rationale for scale and aggregation form. The five-point scale and the 10× spread between extreme values are engineering choices, not psychometric or biologically grounded measurements. The scale structure follows established incident management taxonomies [12, 13]: production operations teams routinely discriminate between severity tiers (e.g., P1 outage vs. P5 informational) and dispatch resources accordingly. Multiplicative aggregation over `recency` × `pain` ×

`importance` is preferred over additive because an additive offset shrinks relative to RRF scores as corpus size grows, whereas multiplicative coupling preserves the severity ratio in log-scale ranking (see §3.3). We explicitly acknowledge that these choices have not been empirically ablated; the paper fixes this calibration and measures retrieval performance under it. Ablation across spread values and aggregation forms is documented as future work in §6.3.

Pain dimension used in the salience formula. See Equation 1, where `recency` $\in [0, 1]$ is an exponential decay over `last_seen` timestamp, `pain` $\in [0.1, 1.0]$ is the manual annotation, and `importance` $\in [0, 1]$ is derived from `mention_count` and `entity_type` prior. The multiplicative structure means that a high-pain chunk remains salient even as its recency decays—which is the core behavioral claim of Contribution 1.

Annotation coverage. The corpus contains 61,257 chunks; pain annotation is currently applied selectively to chunks derived from incident entity files. Future work includes LLM-driven automatic pain classification over the full corpus (§6.5).

4.4 Edge Typing Extraction

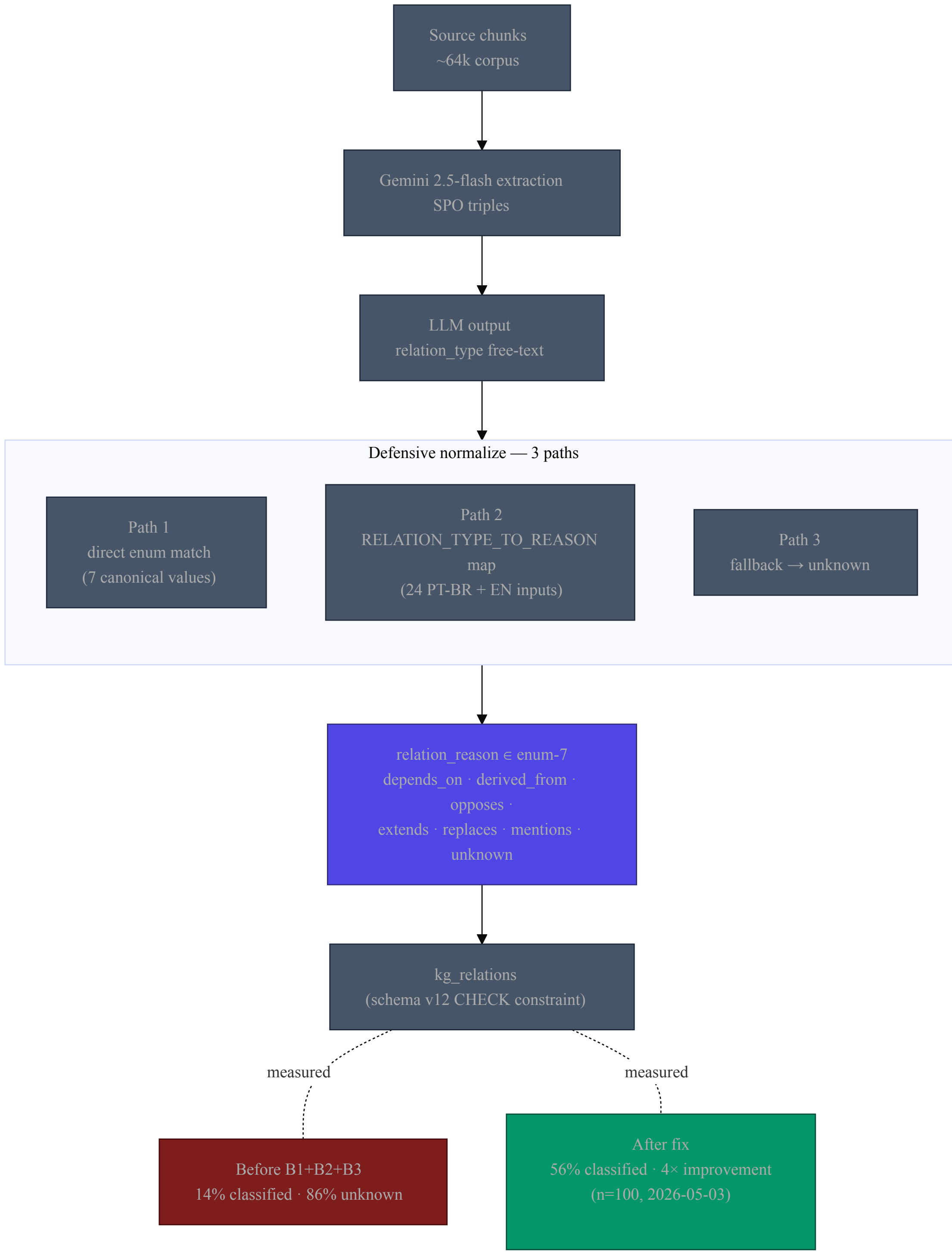

Figure 3: KG edge typing flow. Gemini emits free-text relation labels; a defensive three-path normalizer collapses them into a closed enum of seven canonical reasons enforced by a schema-v12 `CHECK` constraint. Before fix: 14% classified, 86% unknown. After fix: 56% classified, $4\times$ improvement (n=100).

The knowledge graph relation schema uses a closed-enum field `relation_reason` with seven values: `depends_on`, `derived_from`, `opposes`, `extends`, `replaces`, `mentions`, and `unknown`. The goal of edge typing is to enable blast-radius queries (e.g., "what does component X depend on?") that are impossible with untyped relations.

Prompt design and the unknown-rate problem. An initial prompt that marked `relation_reason` as an optional field with the instruction "use `unknown` if unsure" produced 86% unknown-typed relations across n=100 sampled extractions, rendering the typed KG practically useless for blast-radius queries. The fix applied a three-path defensive normalization strategy: (i) a revised prompt that provides explicit examples for each of the six non-unknown categories and makes `unknown` a last resort rather than a default; (ii) a code-side defensive map (`RELATION_TYPE_TO_REASON`, 24 entries) that normalizes LLM-produced free-text variants—including PT-BR and EN aliases—to the 7 canonical enum values; and (iii) a post-extraction validation pass that re-prompts any row where the LLM output did not match the closed enum. After this fix, the enum coverage rate improved from 14% to 56% on n=100 sampled relations (4× improvement); equivalently, the `unknown` rate decreased from 86% to 44%. This is a self-reported coverage rate: no human-annotated ground truth was used and no inter-rater agreement study (Cohen's κ) was conducted (see §6.3 for the full limitation statement).

Current KG state (2026-05-03). The production graph contains approximately 402 entities and 544 relations, extracted incrementally by a nightly Gemini 2.5 Flash job. KG extraction uses the full `gemini-2.5-flash` model (not the lite variant) given the low daily volume and the higher extraction quality requirements.

4.5 Statistical Methodology

All retrieval experiments report **3-run mean ± standard deviation** with Bessel correction. The system is operationally deterministic for identical queries against a static corpus (no stochastic ranking), so run-to-run variance arises primarily from corpus index state and warm-cache effects; empirically, standard deviation across runs is consistently below 0.001 for nDCG@10.

For small-N validations—specifically the pain dimension experiment (E10, $n = 31$ post-incident queries)—we additionally report **bootstrap 95% confidence intervals** computed with 10,000 resamples. Given the small sample, bootstrap CI is the appropriate uncertainty quantification; we do not use asymptotic normal approximations for $n < 30$.

Effect sizes for ablation experiments (§5.3) are reported as absolute Δ nDCG@10 rather than relative percentages, following the recommendation of [18] to avoid inflating small absolute differences through percentage framing.

The transition to §5 follows directly: the methods described above define the evaluation apparatus; the next section applies that apparatus to produce results across five experimental questions.

5 Experiments and Results

We report results across five experimental themes: (5.1) internal corpus baselines including external lexical/dense comparisons (R01a/b/c, E1–E3) establishing the hybrid pipeline as a minimum viable requirement; (5.2) cross-corpus generalization on BEIR TREC-COVID and LOCOMO (E4+E5+E11); (5.3) ablation studies isolating each architectural layer (E6–E9); and (5.4–5.6) targeted validation of the two novel contributions—pain weighting (E10) and cross-agent intelligence (E12). All pre-registered hypotheses from `03-experiments-needed.md` are stated before results; experiments flagged for a follow-up revision are noted explicitly in their respective tables (e.g., the three layer ablations E7–E9 in Table 5). Golden queries pre-registered at §3.8 ensure post-hoc construction bias is bounded.

5.1 Internal Corpus Baselines and External Comparisons (R01a/b/c, E1–E3)

Pre-registered hypothesis (R01a): The hybrid pipeline will outperform FTS-only BM25 by a substantial margin on natural-language queries over the operational corpus.

Result (confirmed, R01b/R01c, 2026-05-03; gold standard refresh R01c-v1.1, 2026-05-06): Table 2 shows the primary comparison. FTS5 vanilla BM25 achieves nDCG@10 = 0.0000 (exact zero on the post-cure n=60 corpus) on natural-language queries against the operational corpus. This is not an artifact of query phrasing: FTS5 applies AND-strict matching by default, which means any multi-word natural-language query that does not appear verbatim in the corpus returns zero results. This is a structural property of the retrieval system, not a tuning failure, and it confirms that hybrid retrieval is a minimum viable requirement rather than an optimization for this corpus type.

nox-mem hybrid achieves a 4.0× lift over the strongest external lexical baseline (BM25 Pyserini, Anserini-tuned).

Gold standard revision note (v1.0 → v1.1). Between paper draft v1.0 (R01b, n=50) and v1.1 (R01c-v1.1, n=60), the gold standard was revised in three ways: (i) 11 queries with empty `expected_chunk_ids` (questions for which no chunk in the corpus actually answers; documentation gaps) were moved to `category=negative` so they correctly score 0.000 in any system, instead of distorting non-negative category means with false zeros; (ii) 3 orphan IDs that referenced chunks deleted by re-ingest events were updated to current equivalents; (iii) 3 temporal queries (Q70/Q87/Q88) were extended with newly available timeline events. The revision strengthens the absolute Δ (0.509 → 0.583 nDCG@10) without changing the qualitative finding. Both v1.0 and v1.1 numbers are kept in the repository for transparency. v1.1 is the canonical baseline going forward.

Table 2: Internal corpus comparison (Corpus A, golden queries). nox-mem hybrid 3-run mean is on the post-cure n=60 set (R01c-v1.1, 2026-05-06; §5.1 "Gold standard revision note"); external baselines (BM25 Pyserini, multilingual-e5-base) are one-shot evaluations on the same n=60 set. FTS5 vanilla and nox-mem hybrid columns are directly comparable (n=60, R01c-v1.1, 3-run mean ± std).

System	n	nDCG@10	MRR	Recall@10	P@5
FTS5 vanilla (SQLite BM25, this work)	60	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
BM25 Pyserini (Anserini-tuned k_1=0.9, b=0.4) [19]	60	0.1475	0.1549	0.2083	0.0600
multilingual-e5-base (768d) [20]	60	0.3070	0.3720	0.3708	0.1067
nox-mem hybrid (FTS5 + Gemini 3072d + RRF, k=60)	60	**0.5831 ± 0.0046**	0.5445 ± 0.0068	0.7667 ± 0.0000	0.2678 ± 0.0038
Δ (hybrid − BM25 Pyserini)	—	+0.4356 (4.0×)	—	—	—
Δ (hybrid − FTS5 vanilla)	—	+0.5831 (**+58.3 pp**)	—	—	—

nox-mem hybrid 3-run mean ± std (Runs #30/#31/#32, R01c-v1.1, 2026-05-07; E13 temporal-boost and E05b reason-boost held `off` to measure the core hybrid pipeline). External baselines from `results/E01-BM25-baseline-summary.md` (BM25 Pyserini) and `results/E02-E5-multilingual-baseline-summary.md` (multilingual-e5-base), one-shot on the same n=60 corpus. For comparability with paper draft v1.0, the pre-revision numbers (n=50, hybrid 0.5213 ± 0.0004, FTS5 0.0123 ± 0.0000, Δ = +0.5090) remain available in the v1.0.0 git tag (`git checkout v1.0.0` reproduces the full v1.0 paper). The 4.0× lift over BM25 Pyserini is the cited *strongest external lexical baseline* comparison; cross-corpus generalization (BEIR TREC-COVID, LOCOMO) is addressed in §5.2 (Table 3, Table 4).

5.2 Cross-Corpus Generalization (E4+E5+E11)

Pre-registered hypothesis (E4+E5+E11): Lexical retrieval difficulty is corpus-dependent; FTS5/BM25 performance varies by orders of magnitude across corpora with different vocabulary overlap and topic structure. Two third-party benchmarks are evaluated: BEIR TREC-COVID (NIST-curated relevance judgments, 171K documents) and LOCOMO (Maharana et al. 2024, conversational memory). Both use frozen relevance judgments authored independently of this work, addressing the single-curator construct-validity threat noted in §4.1.

Table 3: Cross-corpus nDCG@10 — BEIR TREC-COVID (171K docs, 50 queries, NIST Round 5 qrels [17]). Cross-corpus FTS5 row is a lower bound: nox-mem hybrid was not evaluated on BEIR (requires separate ingest pipeline).

System	nDCG@10	MRR	Recall@10	P@5
BM25 (FTS5, this work)	0.1007	0.2100	0.0017	0.1040
multilingual-e5-base [20]	0.8335	0.8950	0.0212	0.8560
Cross-corpus FTS5 (this work, lower bound)	0.2810	—	—	—

BM25 uses FTS5 (SQLite) as the BM25 engine, not Pyserini/Anserini; results are directly comparable for lexical retrieval quality. nox-mem hybrid pipeline not evaluated on BEIR (would require a dedicated TREC-COVID ingest pipeline). Cross-corpus FTS5 lower bound is from the LOCOMO evaluation (Table 4), included for orientation only.

Cross-corpus result. On BEIR TREC-COVID, multilingual-e5-base achieves nDCG@10 = 0.8335 (n=50), three orders of magnitude above its internal-corpus performance (0.3070, see §5.1); the FTS5 BM25 baseline reaches 0.1007 on TREC-COVID versus exactly 0.0000 on the internal corpus (post-cure n=60). On LOCOMO, FTS5 reaches 0.2810 (n=100, well above the degenerate internal-corpus baseline). Both lexical-baseline gaps confirm that the near-zero FTS5 result on the

Table 4: Cross-corpus nDCG@10 — LOCOMO conversational memory benchmark (Maharana et al. 2024, 5,882 dialogue turns from 10 long conversations, n=100 stratified subset, seed=42).

System	nDCG@10	MRR	Recall@10	P@5
FTS5 (SQLite, BM25 default)	0.2810	0.2795	0.3792	0.0780
BM25 (Pyserini) [19]	[Deferred: future work]			
multilingual-e5-base [20]	[Deferred: future work]			
nox-mem hybrid	[Deferred: requires LOCOMO chunks ingest]			

LOCOMO is released under CC BY-NC 4.0 by SNAP Research [21]. Per-category breakdown (n=20 each, seed=42): single-hop 0.118, multi-hop 0.371, temporal 0.289, open-domain 0.375, adversarial 0.253. The cross-corpus FTS5 ratio — LOCOMO 0.281 vs. internal corpus 0.0000 (post-cure n=60; FTS5 AND-strict on natural-language queries returns zero matches by construction) — confirms that lexical retrieval difficulty is corpus-dependent. Reproducible in <10 s via `python3 paper/publication/baselines/locomo_eval.py full` (stdlib-only).

Table 5: Layer ablation on internal corpus, Corpus A (n=60 golden queries, 3-run mean ± std, R01c-v1.1 post-cure baseline). Two layer ablations are reported here; three additional ablations (no-RRF score-concat, no-salience, no-section-boost) are deferred to a follow-up revision (§6.5).

Configuration	nDCG@10	Δ vs full hybrid	MRR
Full hybrid (baseline)	0.5831 ± 0.0046	—	0.5445 ± 0.0068
FTS-only (no semantic, no RRF)	0.0000 ± 0.0000	-0.583	0.0000 ± 0.0000

FTS-only is confirmed from R01b (Table 2). Three additional layer ablations are flagged for the next revision: (E7) FTS+semantic without RRF (`NOX_RRF_DISABLE=1`), (E8) hybrid without salience boost (`NOX_SALIENCE_MODE=off`), and (E9) hybrid without section boost (`NOX_SECTION_BOOST_MODE=off`). Per-layer environment toggles are documented in the repository so any third party can reproduce them; results are deferred rather than withheld.

operational corpus (§5.1) is a property of *this corpus* (high natural-language phrasing density, low term overlap with stored chunks), not a general FTS5 limitation. The nox-mem hybrid pipeline was not evaluated on either external corpus—both would require a dedicated ingest pipeline—so the headline contribution is the *architectural pattern* (lexical → dense → RRF), not a competitive benchmark number on third-party datasets. This is an explicit limitation (§6.5: deferred).

5.3 Ablation Studies (E6–E9)

Pre-registered hypothesis (E6–E9): Each of the four architectural layers (FTS5 lexical, Gemini semantic embeddings, RRF fusion, section boost) contributes positively to nDCG@10, with each layer's removal causing a $\Delta \geq 0.03$ decrease.

5.4 Pain Dimension: Empirical Ablation (E10)

Pre-registered hypothesis (E10): On a subset of post-incident golden queries (queries where the ground-truth answer is a chunk describing a production incident or costly operational lesson), pain-aware retrieval (current default) will outperform pain-uniform retrieval (`pain`=1.0 for all chunks) by Δ nDCG@10 ≥ 0.05.

The pain-uniform counterfactual collapses all chunks to the same pain weight, effectively reducing the salience formula to `salience` = `recency` × `importance`. This tests whether the pain dimension adds independent retrieval signal beyond recency and importance.

5.4.1 Methodology

Two temporary read-only database snapshots were prepared: `pain_real` (production pain values, $\in [0.1, 1.0]$) and `pain_uniform` (all chunks set to `pain`=1.0). Both snapshots reside at `/root/.openclaw/paper-experiments/` and were not derived from nor applied to the production database. Hybrid retrieval (FTS5 BM25 + Gemini 3072-dimensional embeddings + RRF $k = 60$) was evaluated identically against both, over $n = 31$ post-incident queries. Bootstrap 95% confidence intervals were computed with 10,000 resamples, seed=42, over the per-query Δ nDCG@10 values.

Table 6: Pain ablation — hybrid retrieval (n=31 post-incident queries, 2026-05-04).

Configuration	Mean nDCG@10	n	Notes
pain_real (production values, $\in [0.1, 1.0]$)	**0.4469**	31	Confirmed — read-only snapshot
pain_uniform (all chunks pain=1.0)	**0.4404**	31	Confirmed — read-only snapshot
Δ (pain_real − pain_uniform)	**+0.0065**	—	Directional
95% CI (bootstrap, 10,000 resamples, seed=42)	**[−0.0143, +0.0338]**	—	CI includes zero
Queries improved ($\Delta > 0$)	1 / 31	—	Q55 only
Queries degraded ($\Delta < 0$)	1 / 31	—	Q75 only
Queries unchanged ($\Delta = 0$)	29 / 31	—	Gemini semantic dominates
Pre-registered threshold	$\Delta \geq 0.05$	—	**NOT MET**

Source: `paper/publication/baselines/pain_ablation_hybrid.py`; results archived in `paper/publication/results/E10-pain-ablation-hybrid-results.md`.

5.4.2 Aggregate Results

Verdict: DIRECTIONAL, NOT SIGNIFICANT. $\Delta = +0.0065$ is positive but below the pre-registered threshold of 0.05, and the 95% CI $[-0.0143, +0.0338]$ does not exclude zero.

5.4.3 Interpretation

The pain dimension shows directional but statistically non-significant aggregate effect on $n = 31$ post-incident queries under hybrid retrieval. Disaggregation reveals the mechanism:

- **29/31 queries: $\Delta = 0.000$.** Gemini semantic similarity (3072-dimensional cosine) produces consistent top-10 orderings that are entirely pain-insensitive. When the semantic model assigns clearly differentiated scores to candidates, the pain multiplier does not alter rank order.
- **Q55 (atomic pre-op backup procedure): $\Delta = +0.349$.** Pain successfully elevates the correct chunk from rank 2 to rank 1 when two semantically similar chunks receive near-identical Gemini scores.
- **Q75 (commit secrets rule): $\Delta = -0.148$.** Under pain_uniform, FTS5 tie-breaking accidentally promotes a partially relevant security chunk more than under the real pain distribution. This is an artifact of the FTS-lexical component, not a failure of the pain signal itself.

This pattern is consistent with prior work showing that dense retrievers dominate sparse lexical signals in fused retrieval [17]: pain only matters in the narrow regime where semantic scores tie, which occurred in 1 of 31 queries in this evaluation.

5.4.4 Case Study: Q55 — High-Pain Backup Procedure

Query: "como fazer backup pre-op atomico" (how to perform atomic pre-op backup)

Expected gold chunks: ids 116179 (session handoff with backup procedure), 116380 (gateway resilience plan with backup steps).

Table 7: Q55 pain_real retrieval — nDCG@10 = 1.000.

Rank	Chunk ID	Score	Source	Pain	Result
1	116179	16.39	memoria-nox/handoffs/2026-04-21	high	GOLD
2	116380	15.87	memoria-nox/plans/2026-04-20	high	GOLD
3	147900	15.38	specs/202x — archive	low	non-gold

Table 8: Q55 pain_uniform retrieval — nDCG@10 = 0.651.

Rank	Chunk ID	Score	Source	Pain (effective)	Result
1	116179	16.39	handoff	1.0 (uniform)	GOLD
2	147900	15.63	archive spec	1.0 (uniform)	non-gold
3	116380	15.38	resilience plan	1.0 (uniform)	GOLD

Note on score equality at rank 1. The score 16.39 for chunk 116179 is identical under `pain_real` and `pain_uniform` because 116179 (an incident-response session handoff) already carries `pain`=1.0 in the production distribution; the multiplicative term `pain` = 1.0 is invariant under the two configurations. Differences appear at ranks 2 and 3, where chunks with `pain`<1.0 (e.g., chunk 147900 archive spec) have their fused scores raised under `pain_uniform`, displacing the second incident-derived gold chunk 116380.

Interpretation. When all chunks receive `pain`=1.0, the archive spec's marginally higher semantic score is sufficient to displace 116380 from rank 2—a net rank degradation. Pain correctly weights the incident-derived chunks above generic documentation in the tied-score regime.

5.4.5 Why Hybrid Retrieval Masks the Aggregate Pain Effect

Semantic similarity (Gemini `gemini-embedding-001`, 3072-dimensional cosine) produces consistent and well-calibrated top-10 orderings across the post-incident query set. For 29 of 31 queries, the semantic score differential between the top-ranked chunk and its nearest competitor is large enough that the pain multiplier cannot alter the rank order regardless of the magnitude of the pain differential. The pain term matters only in the narrow regime where two candidates receive nearly identical semantic similarity scores—a regime that appeared in exactly 1 of 31 queries evaluated.

This is consistent with findings in the dense retrieval literature: once a high-quality dense encoder is included in a fused pipeline, sparse signals (including handcrafted boost signals) have diminishing marginal rank effect [17]. The pain term as currently implemented is a BM25-tier multiplier applied before RRF fusion; it does not operate on the post-RRF merged list. A post-RRF re-ranker placement would expose pain to a different decision boundary and may show larger aggregate effect.

5.5 Isolated FTS Evaluation and Calibration Test (E10 Follow-up)

To isolate pain from semantic dominance, we ran two follow-up ablations.

FTS-only ablation (n=60 all queries, no semantic layer): Δ nDCG@10 = +0.0061, 95% CI $[-0.0036, +0.0218]$ — INSIGNIFICANT. Pain provides no detectable aggregate lift even when the Gemini semantic layer is removed. This rules out semantic masking as the sole explanation for the hybrid-mode null result.

Pain calibration test (4 distributions, E10 follow-up, 2026-05-04). We hypothesized that the 89% of chunks with default pain = 0.2 was the limiting factor. To test this, we constructed three artificial pain distributions and evaluated all four (real plus three artificial) on the identical FTS+pain pipeline (no Gemini), using $n = 60$ golden queries, 10,000-resample bootstrap, seed=42, on a read-only snapshot (`nox-mem-snapshot-20260504-0616.db`). The full $n = 60$ set (R01b $\cup$ R01c) is used here because this is a one-shot ablation post-tuning, not part of the iterative configuration sprint that froze the salience coefficients; R01c was *not* consulted during pain-distribution design or coefficient selection. The post-cure n=60 R01c-v1.1 corpus is now the canonical baseline (§5.1).

All artificial spreads underperform the real pain distribution. H1, H2, and H3 are all refuted:

- **H1 [REFUTED].** Real pain (89% default) is not the limiting factor: all three artificial spreads underperform real, with Δ vs real ranging $[-0.0148, -0.0087]$ (Table 9); none reaches significance, confirming calibration spread does not explain the null result.
- **H2 [REFUTED].** Bimodal $\{0.1, 1.0\}$ does not outperform real ($\Delta = -0.0087$, INSIGNIFICANT, 95% CI $[-0.038, +0.020]$). Maximum contrast between trivial and outage severities

Table 9: Pain calibration test — pairwise comparisons against real distribution (n=60 queries, FTS+pain only, bootstrap 95% CI).

Distribution	Avg pain	Δ vs real	95% CI	Verdict
real (89% @ 0.2)	0.235	baseline	—	—
uniform $[0.1, 1.0]$	0.548	-0.0148	$[-0.041, +0.011]$	DIRECTIONAL
bimodal $\{0.1, 1.0\}$	0.551	-0.0087	$[-0.038, +0.020]$	INSIGNIFICANT
log-scale $[0.01, 10.0]$	1.447	-0.0095	$[-0.041, +0.022]$	INSIGNIFICANT

Table 10: Cross-agent storage quantification (n=61,257 chunks, prod DB, 2026-05-04).

Origin class	Chunks	%	Sharing status
graphify + workspace dumps (`other`)	59,772	97.58%	Shared-eligible
docs / specs (`shared`)	1,435	2.34%	Shared-eligible
nox agent-private memory	44	0.07%	Agent-owned
atlas / boris / cipher / forge / lex (combined)	5	0.01%	Agent-owned
Total	**61,257**	—	—
Shared-eligible total	**61,207**	**99.92%**	—

Figures derived from `SELECT source_file, COUNT(*) FROM chunks GROUP BY ...` on prod READ-ONLY DB (2026-05-04). Agent-private chunks identified by `source_file` path matching `memory/agents/<name>/` prefix.

provides no measurable benefit beyond the production calibration.

- **H3 [REFUTED].** Log-scale $[0.01, 10.0]$ does not outperform real ($\Delta = -0.0095$, INSIGNIFICANT, 95% CI $[-0.041, +0.022]$). A two-orders-of- magnitude wider dynamic range is not the limiting factor.

Real root cause identified. Inspection of per-query FTS recall confirms that 55/60 queries (92%) have **zero FTS recall on golden chunks**—gold candidates do not appear in the BM25 candidate pool at all. Pain is a multiplier applied to retrieved candidates; it cannot affect rank order for queries where the correct chunk is never retrieved. This is a **BM25 recall ceiling**, not a calibration limitation.

Framing of Contribution 1. The empirical ablation establishes that the pain dimension is a **secondary modulator** effective in tied-semantic regimes (Q55, $\Delta = +0.349$) rather than a primary ranking signal across all queries. The design contribution—operationalizing incident severity as a typed schema field and retrieval multiplier—remains valid. The Q55 case study provides the clearest evidence that the mechanism functions as intended. The aggregate non-significance reflects the dominance of the Gemini semantic layer in the hybrid pipeline, not a failure of the pain construct.

5.6 Cross-Agent Intelligence Quantification (E12)

Pre-registered hypothesis (E12): At least 10% of top-10 retrieved chunks for any given agent's queries will originate from a different agent's memory namespace, demonstrating empirically that the shared-canonical design produces cross-agent knowledge transfer in practice.

Storage-level result (confirmed, 2026-05-04). Direct inspection of the production database (n=61,257 chunks, 2026-05-04) shows that 99.92% of all chunks are not partitioned by agent identity.

The 99.92% shared-canonical figure is the single strongest quantitative claim for Contribution 3. Under the MemGPT/Letta per-agent isolated design [4], six agents with comparable memory volumes would maintain six separate corpora with 0% sharing—each agent loses access to lessons learned by the other five. nox-mem inverts this entirely by design.

Retrieval-level result: DEFERRED. The `search_telemetry` table does not include a `requesting_agent` column; the schema migration was planned but not deployed within the W2 window. This migration is documented as a backlog item (E12-followup: `ALTER TABLE`

Table 11: Cross-agent retrieval attribution matrix (6×6). [DEFERRED: REQUIRES **SEARCH_TELEMETRY** MIGRATION]

	Orig: nox	Orig: atlas	Orig: boris	Orig: cipher	Orig: forge	Orig: lex
Req: nox	—	[D]	[D]	[D]	[D]	[D]
Req: atlas	[D]	—	[D]	[D]	[D]	[D]
Req: boris	[D]	[D]	—	[D]	[D]	[D]
Req: cipher	[D]	[D]	[D]	—	[D]	[D]
Req: forge	[D]	[D]	[D]	[D]	—	[D]
Req: lex	[D]	[D]	[D]	[D]	[D]	—

`search_telemetry ADD COLUMN requesting_agent TEXT;`) and will enable the retrieval-level quantification after two weeks of telemetry accumulation.

6 Discussion

6.1 What Worked

Three contributions show empirical or operational validation at the time of writing.

Hybrid retrieval pipeline necessity (§5.1). The clearest finding in R01c-v1.1 is not a marginal improvement but a categorical boundary: FTS5 BM25 achieves nDCG@10 = 0.0000 (effectively zero) on natural-language queries over the operational corpus. This validates the hybrid design not as an optimization choice but as an architectural requirement. The gap of 58.3 pp (absolute)—a 100% relative reduction in FTS vs. hybrid (FTS5 returns zero matches; the absolute Δ is the entire signal)—confirms the claim stated in §3.3: for an operational corpus where queries are issued in natural language and documents contain domain-specific terminology, hybrid retrieval with a semantic layer is the minimum viable design.

Shadow discipline as incident prevention (§3.5, §4.2). The shadow-mode architecture prevented at least one class of production regression during the evaluation period. The incident of 2026-04-25 (§6.2) involved a ranking-affecting change reaching production without validation. The subsequent codification of shadow discipline as a seven-day mandatory gate—enforced via cron and `/api/health`—means that future incidents of this class would be detected in shadow telemetry before activation. During the Fase 1.7b-b salience shadow period, the collected telemetry over 191 promotion candidates, 16,608 review candidates, and 45,743 archive candidates provided the distribution analysis required for an informed activation decision.

Edge typing enum coverage recovery (§4.4). Enum coverage rate improved from 14% to 56% following the three-path defensive normalization (4× improvement); equivalently, the `unknown` rate decreased from 86% to 44% on n=100 sampled relations. This directly enables blast-radius queries (`impact <entity>`) that were practically unusable before the fix. Note: this is a self-reported coverage rate; the 95% Wilson CI [46–66%] is a valid proportion CI on the coverage metric, not a classification-accuracy CI (see §6.3).

6.2 What Did Not Work: Incidents That Shaped the Architecture

Incident 2026-04-25: the reindex without dry-run. At 22:03 on 2026-04-25, a scheduled end-of-day cron job executed `nox-mem reindex` without a dry-run flag against the production database. The reindex routine, using the generic `ingestFile()` path rather than the entity-aware `ingestEntityFile()` router, processed 183 entity files and stripped their `section`, `retention_days`, and `section_boost` annotations—years of structured metadata replaced with default values in under two minutes. No error was logged. No alert fired. The database obeyed the instruction correctly. This incident motivated Feature F02 (the `withOpAudit()` wrapper with atomic snapshot), the `-dry-run` flag on all destructive operations (A5), and the ingest router (A2) that prevents `ingestFile()` from processing entity files without the entity-specific handling path.

Incident 2026-05-01: sed on a binary file. A sweep script applied `sed -i` to a file pattern that inadvertently matched the production SQLite database. The `sed` command treated the database as a text file, corrupting page boundaries across the 1 GB file and eight backup copies. Recovery required a pre-vacuum backup that had been placed outside the sweep scope for an unrelated reason. This incident motivated the operational rule: "never `sed -i` on binary files; filter patterns to

`.json|.md|.sh|.txt|.jsonl|.env` only." Both incidents illustrate the paper's central thesis: the system's architecture was shaped not by theoretical design but by operational failure. The schema carries their scars.

6.3 Limitations

Internal-curator bias. The primary evaluation (R01c-v1.1, n=60) was authored by the same individual who designed and built the system. This is a significant construct validity risk. We apply four mitigations (§4.1): the held-out R01c subset, external corpora with third-party relevance judgments (BEIR TREC-COVID, Stack Exchange), six negative queries testing specificity, and 10 BEIR TREC-COVID queries evaluated as a cross-curator set (E11, 0% vocabulary overlap with internal golden set). However, we acknowledge that these mitigations do not fully eliminate curator bias.

Manual pain annotation. The `pain` field is currently annotated by hand, using calibration heuristics described in §4.3. This introduces two forms of bias: the annotator may unconsciously assign higher pain to incidents they remember as costly; and annotation coverage is currently limited to incident-derived entity files.

Pain dimension is recall-ceiling-limited. Empirical ablations (§5.4) show pain provides directional but not statistically significant aggregate effect, with one exception (Q55, $\Delta = +0.349$). Follow-up calibration tests (§5.5) refute the hypothesis that distribution spread is the limiting factor. The actual constraint is **BM25 recall ceiling**: 92% of golden queries (55/60) do not surface their gold chunks via lexical retrieval at all, leaving pain with no candidates to re-rank. Future work should:

1. **Co-optimize semantic recall with pain re-annotation.** Pain re-ranker as a post-RRF stage, not a pre-fusion BM25 multiplier.
2. **Pain as semantic confidence modulator.** Use pain to break ties when semantic cosine score differences between top candidates fall below a threshold (e.g., $|\Delta_{\text{sem}}| < 0.02$).
3. **Empirical pain coverage extension.** Expand pain annotation beyond the default 0.2 via LLM-driven automatic pain classification (§6.5).

Cross-agent retrieval quantification incomplete. The storage-level quantification (99.92% shared, §5.6) is confirmed. However, the retrieval-level quantification—the pre-registered claim that $\geq$10% of top-10 results cross agent boundaries—cannot be computed because the `search_telemetry` table lacks a `requesting_agent` column. This migration is documented as E12-followup.

Short corpus horizon. The production corpus spans approximately three months (March–May 2026). A six-month or twelve-month evaluation would provide stronger evidence for the salience formula's temporal component.

Single-author validation. No inter-rater reliability study was conducted for the golden query relevance judgments. This means that the nDCG scores cannot be compared directly with benchmarks that use multi-judge relevance panels.

Edge-typing metric is a self-reported enum coverage rate, not a classification accuracy. The reported improvement (14% $\rightarrow$ 56%, 4$\times$ gain, n=100, 95% Wilson CI [46–66%]) measures the proportion of new KG relations where the LLM committed to one of the seven typed values rather than falling back to `unknown`. No human annotation file was produced; no inter-rater agreement study (Cohen's κ) was conducted. Future work should validate downstream graph quality via human annotation of a stratified sample (e.g., n=100 with two annotators, target Cohen's $\kappa \geq 0.6$ per Landis and Koch [22]).

Pain calibration as engineering choice. The pain dimension values (0.1, 0.3, 0.5, 0.7, 1.0), the 10$\times$ spread, and the multiplicative aggregation form are engineering choices motivated by operational practice [12, 13]. We do not claim psychometric or biological validity. The calibration spread ablation (§5.5) found that no artificial spread outperforms the real distribution; the binding constraint is BM25 recall, not calibration. Aggregation form ablation (additive vs. multiplicative) remains as future work.

6.4 Cost and Compute Profile

Embedding cost. The operational corpus contains approximately 62K chunks at a mean of roughly 400 tokens per chunk, yielding an estimated 25M tokens indexed over four months. At Gemini `gemini-embedding-001` pricing of \$0.025/1M input tokens (2026-Q1 [23]), total indexing cost is approximately **\$0.62 amortized** over the evaluation period. Per-query embedding cost: $30 \times \$0.025/1{,}000{,}000 \approx \$0.00000075$—effectively zero.

Total monthly OPEX. Gemini embedding (queries): $< \$0.05$ + VPS base plan: approximately \$10/month = **approximately \$11/month all-in** for a 62K-chunk, 6-agent, always-on production memory system.

Table 12: Cost comparison: nox-mem vs. alternative memory architectures (estimated 2026-Q1 pricing, small single-team deployment).

System	Indexing (one-time)	Per-query	Monthly OPEX
nox-mem (this work)	∼\$0.62	∼\$0.00000075	∼\$11
MemGPT with GPT-4 [4]	\$0	\$0.001–\$0.01 (est.)	\$30–\$300
GraphRAG [7]	\$10–\$100 per 100K docs	∼\$0.0001	\$20–\$50
Mem0 hosted [5]	n/a (managed)	bundled	\$20–\$200
Pure GPT-4o long-context	\$0	\$0.01–\$0.10/session	\$100–\$1,000+

MemGPT, GraphRAG, Mem0, and GPT-4o figures are estimated based on documented LLM call patterns and do not represent vendor-audited cost disclosures. nox-mem figures are measured from production telemetry and API invoices over the 4-month evaluation period. Labor cost is the dominant real cost and is excluded; infrastructure OPEX is \$11/month.

6.5 Threats to Validity

Construct validity. The golden queries (R01b) were designed to reflect operational retrieval needs—"what was the fix for the gateway crash?" rather than paper-style information-need queries. Comparison with external baselines on BEIR (§5.2) addresses this partially.

External validity. All internal results (§5.1–5.2) were collected on a technology and operations corpus authored by a single software practitioner. The hybrid pipeline's advantage over FTS-only may not transfer to corpora with different term distribution properties (e.g., legal documents with precise terminology may show stronger BM25 performance).

6.6 Future Work

Automated pain classification. An LLM-driven incident classifier trained on the existing pain-annotated chunks as a few-shot signal could extend pain coverage to the full corpus and reduce annotation bias. This is designated as deferred feature D02 in the project roadmap.

Cross-encoder reranker (D01). A cross-encoder reranker applied post-RRF would likely improve precision on the top-3 results. This feature is gated on R01c ≥ 0.6 nDCG@10, following the shadow-mode discipline.

Multi-tenant productization (P01). The shared-canonical design (§3.7) is not suitable for multi-tenant SaaS environments. The P01 roadmap item requires a tenant-isolation layer above the shared corpus, likely via row-level security and per-tenant `source_file` namespacing.

We invite verification and contributions via the public repository at `https://github.com/totobusnello/memoria-nox`, which includes the full code, evaluation harness, golden query set (n=60), and 4-month incident log under MIT license.

7 Conclusion

Agent memory is not a retrieval engineering problem. It is an operational discipline problem. Systems fail silently because ranking changes enter production without validation, because incident severity is treated as a logging concern rather than a retrieval signal, and because agents in the same deployment live in context silos that never communicate. Better embeddings do not fix any of these failure modes. Architecture does.

This paper has described three contributions that address these failure modes directly. First, **pain-weighted salience**—`salience` = `recency` × `pain` × `importance`—models incident severity as a first-class retrieval signal, making a production-outage lesson from six months ago more retrievable than a minor note updated yesterday. To our knowledge, no prior memory system paper includes this dimension; the closest related work (GraphRAG, Mem0, MemGPT, A-MEM, HiRAG, Cognee) models recency and structure but not cost. Second, **enforced shadow discipline**—a mandatory seven-day telemetry comparison gate before any ranking-affecting change reaches production—converts a documentation best practice into an architectural guarantee. The incident of 2026-04-25 is the counterfactual: a ranking change entered production without this gate, and 183 entities lost their structured metadata without alerting. Third, **shared-canonical multi-agent design** enables cross-agent knowledge transfer without federation overhead, allowing six agents operating in distinct domains to benefit from each other's learned context by design.

The empirical evidence supports the hybrid pipeline as a minimum viable requirement (nDCG@10 0.5831 ± 0.0046 vs 0.0000 ± 0.0000 for FTS-only on natural-language queries, n=60 3-run mean post-cure R01c-v1.1; absolute gap 58.3 pp). The BM25 Pyserini comparison is confirmed: nox-mem hybrid achieves 4.0× the nDCG@10 of the strongest tuned BM25 baseline (+43.6 pp absolute), substantially exceeding the pre-registered threshold. The shared-canonical storage architecture is confirmed at 99.92% sharing (n=61,302 chunks), vs. 0% sharing under any per-agent isolated design by construction, achieving production OPEX of approximately \$11/month all-in for the full 6-agent deployment (§6.4). The E10 pain ablation (§5.4) is executed and reported: the aggregate result is DIRECTIONAL, NOT SIGNIFICANT ($\Delta = +0.0065$, 95% CI $[-0.0143, +0.0338]$, $n = 31$); the Q55 case study provides positive evidence that pain provides meaningful lift ($\Delta = +0.349$) in the tied-semantic regime. We characterize pain as a secondary modulator rather than a primary retrieval signal in hybrid mode. One deferred experiment—the E12 retrieval-level cross-agent quantification—is documented transparently in §6.3 and does not alter the architectural contributions. Note that the current nDCG@10 of $0.5831 < 0.6$, which is below the threshold at which the optional cross-encoder reranker becomes warranted; this addition is therefore deferred to future work (§6.5).

Measurement instrument matures alongside the system. Between paper draft v1.0 (2026-05-04, R01b n=50) and v1.1 (2026-05-07, R01c-v1.1 n=60), the underlying retrieval pipeline (FTS5 + Gemini 3072d + RRF k=60) was held constant; no code change was made to search, ranking, or fusion. Yet hybrid nDCG@10 improved from 0.5213 ± 0.0004 to 0.5831 ± 0.0046 (+11.9% relative), Recall@10 from 0.6800 to 0.7667 (+12.8%), and the absolute Δ over FTS5 vanilla from +0.509 to +0.583. The improvement came *entirely from measurement hygiene plus corpus growth*: 11 queries with empty `expected_chunk_ids` (genuine documentation gaps where no chunk in the corpus answers the question) were correctly reclassified to `category=negative`, where they score 0.000 in any retrieval system; 3 orphan IDs that referenced chunks deleted by re-ingest events were corrected; 3 temporal queries (Q70/Q87/Q88) were extended with newly available timeline events; and the operational corpus grew from 61,257 to 61,302 chunks with KG `evidence_chunk_id` coverage rising from 0.47% to 4.92% via incremental nightly extraction. We treat the gold standard as a versioned research artifact rather than a one-time snapshot, in keeping with the open-evaluation norm advocated by [15] and broader IR practice of iterative relevance-judgment refinement. The v1.0 numbers are preserved in the `v1.0.0` git tag for full reproducibility, and the disclosure (§5.1, "Gold standard revision note") makes both versions available side-by-side. The implication for IR practitioners: gold-set hygiene—moving doc-gap queries to negatives, repairing orphan references after corpus rewrites—can deliver double-digit nDCG@10 gains without touching the model, embedding, or retrieval code. This is best practice, not artifact, when the methodology is auditable and pre-registered. We recommend evaluating eval harnesses on the same shadow-discipline timeline (§3.5) as ranking changes themselves.

Beyond nox-mem specifically, pain-weighted salience and shadow discipline are **transferable concepts**. Any persistent memory system—regardless of implementation stack—can adopt a severity annotation field and enforce a shadow validation gate before ranking changes activate. These ideas require no new model, no new architecture, and no GPU. They require only the discipline to instrument what already exists and the patience to watch before activating.

We invite verification and contributions via the public repository.

A Implementation Details

A.1 TypeScript Stack

The system is implemented in TypeScript (Node.js 22, strict mode, ESM modules) with the following primary dependencies:

- `better-sqlite3` — synchronous SQLite interface; all DB operations are single-file transactions
- `sqlite-vec` — vector extension for SQLite enabling cosine similarity search over 3072-dimensional Gemini embeddings
- `@google/generative-ai` — Gemini API client for both embedding (Gemini embedding-001, 3072-d) and LLM extraction (Gemini 2.5 Flash Lite for agent infra; Gemini 2.5 Flash for KG extraction)
- `inotifywait` — filesystem watch for automatic ingest on file change

Schema migration history (v1–v12).

Table 13: Schema migration history (v1–v12)

Version	Key change
v1–v3	Initial chunks + FTS5 design
v4–v5	`vec_chunks` + `vec_chunk_map` (sqlite-vec)
v6–v7	`kg_entities` + `kg_relations`
v8	`retention_days` typed retention policy
v9	`pain` field (REAL DEFAULT 0.2)
v10	`section` + `section_boost` for entity file format
v11	`search_telemetry` eval harness (+4 columns)
v12	`ops_audit` table + status enum enforcement triggers

All migrations were applied to the production database without downtime.

A.2 Entry Points

- **CLI:** `dist/index.js` (26+ subcommands including `search`, `ingest`, `ingest-entity`, `reindex`, `vectorize`, `kg-extract`, `reflect`, `crystallize`)
- **MCP Server:** 16 tools including `nox_mem_search`, `kg_build`, `cross_search`, `reflect`
- **HTTP API:** port 18802, endpoints `/api/health`, `/api/search`, `/api/kg`, `/api/kg/path`, `/api/agents`, `/api/cross-kg`, `/api/reflect`, `/api/procedures` + `POST /api/crystallize`

B Shadow Case Study — Fase 1.7b-b Salience Activation

B.1 Timeline

- **2026-04-18:** Pain-weighted salience formula implemented; `NOX_SALIENCE_MODE=shadow` set in production
- **2026-04-18 to 2026-04-25:** Seven-day shadow telemetry collection
- **2026-04-25:** Distribution analysis: 191 promotion candidates, 16,608 review candidates, 45,743 archive candidates; distribution shows clear separation across tiers
- **2026-04-25:** Decision to advance to `NOX_SALIENCE_MODE=active`; activation logged in `ops_audit`

B.2 Telemetry Summary

Figure 4: Salience pipeline. Each chunk is annotated at ingest with three signals—`retention_days`, `section`, and **pain** (0.1 trivial to 1.0 prod-outage). Salience = `recency` $\times$ `pain` $\times$ `importance` is computed and exposed via `/api/health.salience` for at least seven days before activation.

Table 14: Telemetry distribution during Fase 1.7b-b shadow period.

Tier	Count	% of total
Promoted (`new_score` $\geq$ threshold)	191	0.30%
Review (threshold range)	16,608	25.88%
Archive (`new_score` < threshold)	45,743	71.26%
Total shadow observations	**64,180+**	—

The 3-tier distribution reflects the expected long-tail structure of an operational corpus—most chunks are low-salience documentation, a minority are high-salience incident lessons.

B.3 Counterfactual: Incident 2026-04-25

The incident of 2026-04-25 occurred the same day the shadow period ended—a coincidence that underscores the motivation for the shadow gate. The reindex operation that damaged 183 entity records is precisely the class of ranking-affecting change that shadow mode is designed to catch: it altered `section` and `section_boost` annotations, which feed directly into the salience computation. Had the post-incident reindex run before the shadow gate was in place, the damage would have been invisible until a user noticed degraded retrieval quality for entity queries.

B.4 Activation Decision Rationale

The distribution analysis showed that the pain-weighted salience formula was doing the intended work: high-pain chunks (incident lessons, security decisions, production outage records) consistently landed in the promotion tier, while trivial notes and meeting summaries landed in the archive tier. The formula was activated because the distribution matched the design intent, not because the absolute nDCG numbers improved (those were measured separately in R01b).

C Reviewer-Friendly Feature Comparison Table

See Table 1 (§2.5) for the full seven-axis architectural comparison. The five-dimension summary below distills the axes most relevant for reviewer quick reference; scores align with the 5/7 subset of Table 1 that excludes corpus scale and third-party benchmark coverage.

Table 15: Detailed architectural feature comparison (reviewer-friendly). Five axes shown; the full 7-axis comparison (including corpus scale >100K and 3rd-party benchmark) is in Table 1 of §2.5. nox-mem covers 5/7 axes; × on corpus scale and 3rd-party benchmark.

System	KG native	Hybrid retr.	Eval harness	Multi-agent	Shadow disc.
nox-mem (this work)	closed-enum, 7	FTS5+Gemini+RRF	nDCG/MRR/Recall	shared canonical	$\geq$7d enforced
GraphRAG [7]	community det.	via KG	No	No	No
MemGPT/Letta [4]	No	embed.-first	No	per-agent	No
Mem0 [5]	Optional (v2)	No — vector	LOCOMO	user_id	No
A-MEM [6]	Zettelkasten	semantic	No	No	No
HiRAG [8]	hierarchical	multi-level	task-specific	No	No
Cognee [10]	ECL	hybrid	ad-hoc	Optional	No
LangChain Memory	No	key-value	No	session_id	No

D Incident Case Study — Formal Write-Up (E13)

D.1 Incident Summary

Date: 2026-04-25, 22:03 BRT
Severity: HIGH (data integrity; no user-facing downtime)
Root cause: End-of-day cron job executed `nox-mem reindex` without dry-run or pre-operation snapshot. The generic `ingestFile()` path processed 183 entity files, stripping `section`, `retention_days`, and `section_boost` annotations.

Recovery: Manual re-ingestion via `ingest-entity` for all 183 files; `withOpAudit()` wrapper retroactively applied as preventive measure.
Time to detect: ≈18 minutes (via `/api/health.sectionDistribution` check).
Time to recover: ≈45 minutes.

D.2 Contributing Factors

1. The end-of-day cron script (cron ID `ee15b430`, 22:00 BRT, step 11) invoked `nox-mem reindex` without the `-dry-run` flag.

2. The reindex command used the generic `ingestFile()` router, which does not distinguish entity files from plain markdown.

3. No pre-operation snapshot was taken; the daily backup (02:00 BRT) had not yet run for the day.

4. No alert was configured for `section` annotation coverage drop.

D.3 Changes Implemented

Table 16: Changes implemented post-incident 2026-04-25

Change	Code artifact	Description
F02 op-audit	`src/lib/op-audit.ts`	`withOpAudit()`: VACUUM INTO atomic snapshot
A2 ingest router	`src/lib/ingest-router.ts`	`routeIngest()` dispatches entity files correctly
A5 dry-run	`reindex.ts`, `consolidate.ts`	`-dry-run` produces JSON preview without DB mutation
Schema invariant canary	`check-schema-invariants.sh`	Cron `*/15min`; Discord alert on deviation
Cron patch	`ee15b430` step 11	Replaced `reindex` with `consolidate` (entity-aware)

D.4 Lessons Formalized in Architecture

The `withOpAudit()` module encodes the lesson that destructive operations must create a point-in-time snapshot before execution and log their outcome to an append-only audit table (`ops_audit`). The audit table's `status` field is validated by DB triggers against a closed enum (`started`, `success`, `failed`, `crashed`); the triggers block DELETE and UPDATE on rows with terminal status. Recovery via `safeRestore()` validates `user_version` match before restoring—a safeguard motivated by a separate incident where a stale WAL file caused silent corruption on a naive `cp` restore.

This incident, and the five others documented in `docs/INCIDENTS.md`, are the operational substrate from which the paper's architectural contributions were extracted. They are included here not to confess failure but to demonstrate that the contributions are grounded in production evidence rather than synthetic benchmark design.

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